

Distinguishing Visual Fatigue, Mental Workload and Acute Stress in Immersive Virtual Reality with Physiological Data: pre-test results

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ABSTRACT

Virtual Reality Induced Symptoms and Effects (VRISE) can arise. Experimental paradigms are heterogeneous to assess them, and various factors can induce physiological variations. Therefore, we developed a Stroop task to study and distinguish VRISE. We use eye tracking, ECG, EDA, and VRSQ, NASA-TLX, and STAI-6 questionnaires. Pre-tests have been conducted with 6 subjects exposed to 4 experimental conditions: control, dual task, stressful, and stereoscopy. Subjects report different subjective visual fatigue and mental workload but not stress between conditions. Several physiological features are different between conditions. A VRISE detector can be envisioned based on physiological data and questionnaires as an index.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—Laboratory experiments

1 INTRODUCTION

Immersive Virtual reality (VR) provokes side effects [8] known as Virtual Reality Induced Symptoms and Effects (VRISE). Research is still needed to comprehend VRISE better. Eye-tracking and pupilometry (ET), electrocardiography (ECG), and electrodermal activity (EDA) are popularizing to measure VRISE. Immersing humans in VR implies to encompass various human factors issues as it seems that visual fatigue, acute stress, and mental overload could have some tangles [7]. We report pre-test results to distinguish visual fatigue, acute stress, and mental workload based on physiological data. Experimental results can be fuzzy at explaining which factors and stimuli originate psychophysiological variations used to measure VRISE. Therefore, at least measuring possible alternative effects is interesting [8]. Most previous contributions use non-standardized stimuli to assess VRISE and mainly concentrate on visually-induced motion sickness when using physiological sensors. From our perspective, tested applications are too complex. It makes it hard to generalize results based on physiological measurements. Reducing noise from complex tasks and allowing the progressive introduction of the various factors influencing human psychophysiological responses in VR could be handy. It encouraged us to create a Stroop task-based VR experiment to assess the possible differences between visual fatigue, acute stress, and mental workload and better assign physiological variations to each of them.

Hypotheses: H1: Subjects report different subjective visual fatigue, stress, and mental workload between conditions. H2: Physiological variations are different between visual fatigue, stress, and mental workload conditions.

2 MATERIAL AND METHODS

Participants: 6 subjects (5 men, 1 woman) aged between 21 and 31 years old, $M = 24.16 \pm 4.12$. Two wore glasses. All were researchers or engineering students, very accustomed to VR.

Software and procedure: We used Unity 3D version 2019.4.22f1 to develop the StroopVR application. It consists of 1) a baseline measurement for 5 minutes consisting of a black cross in the center and a grey background; 2) Instructions for a Stroop task: participants are asked to announce out loud the font color by inhibiting the word itself which signifies a color; the Stroop task (25 congruent words, 25 incongruent words), 3) three questionnaires to assess stress, mental workload, and visual fatigue; 4) Instructions to relax; 5) a biophilic relaxation environment; 6) repeating 2) to 5) until all four experimental conditions are done; 7) 5 minutes "Endline" similar to 1). Subjects execute the Stroop task via voice interaction, and interaction in menus is done via combining eye gaze and the controller's trigger.

Hardware: StroopVR is deployed on the HTC Vive Pro Eye. SteamVR assures the links between hardware and software. The physiological sensors used are Shimmer3 ECG unit @256Hz¹; Shimmer3 GSR+ unit @128Hz²; Tobii eye tracker directly integrated into the HTC Vive Pro Eye @90Hz³. The computer used to run the StroopVR application and collect data used Microsoft Windows 10 Business edition, Processor: i9-10980K @ 2.40 GHz; CPU: 32 GB (RAM); GPU: Nvidia GeForce RTX2080 Super.

Dependant variables: 1) Features extracted from ET, ECG, and EDA signals; 2) Scores from three questionnaires VRSQ (cyber-sickness and visual fatigue) [2], STAI-6 (stress) [5], NASA-TLX (mental workload) [1].

Independent variables: Four experimental conditions are compared. 1) *Control*: Stroop task only. No stereoscopy is displayed; 2) *DualTask*: Stroop task + press the trigger on the controller when the word "red" is announced in audio, the audio appears randomly 25 times with 10 occurrences of the word "red." No stereoscopy is displayed; 3) *Stressful*: Stroop task + alarm sound randomly emitted at about 70 dB 10 times with a random duration of 1, 2 or 3 seconds. No stereoscopy is displayed; 4) *Stereoscopic*: Stroop task with negative parallax. We apply a 0.25m parallax (corresponding to the comfortable viewing limit according to [6] translated in SteamVR and Unity 3D) applied on the words. The purpose is to favor a vergence-accommodation conflict that results in visual fatigue [7].

Data pre- and processing: We applied a 50Hz low-pass filter and a Discrete wavelet transform for the ECG signal. We applied a 1Hz low-pass filter and a deconvolution algorithm for the EDA signal. ECG and EDA features have been extracted using Neurokit2 [4]. We applied the Velocity-Threshold Identification fixation classification and the algorithm proposed by [3] for the ET signal. For all signals, we used a 1 minute or 30 seconds window to extract temporal features. We present statistical tests comparing experimental conditions

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¹<http://www.shimmersensing.com/products/shimmer3-ecg-sensor>

²<http://www.shimmersensing.com/products/shimmer3-wireless-gsr-sensor>

³<https://www.vive.com/eu/product/vive-pro-eye/specs/>

during the Stroop task. To test feature differences between conditions, we used rANOVA. For questionnaires scores we used a chi square test. The significance level is tested with a confidence of $\alpha = 0.05$.

3 RESULTS

VRSQ scores are statistically different: Stereoscapy $\neq$ Control $p = < 0.001$, Stereoscapy $\neq$ Stressful $p = 0.003$, Stereoscapy $\neq$ DualTask $p = 0.0002$. NASA-TLX scores are statistically different: DualTask $\neq$ Control $p = 0.002$, DualTask $\neq$ Stressful $p = < 0.001$, DualTask $\neq$ Stereoscapy $p = 0.001$. However, STAI-6 scores are not statistically different between conditions. We tested 133 features to find which are different between conditions. Results (only those statistically significant) are presented in Table 1.

4 DISCUSSION AND LIMITATIONS

H1 is partially supported as subjects do not report different subjective stress between conditions. H2 is supported as several physiological features are different between conditions. Our results show that more data are necessary to distinguish VRise better and that only a few features vary differently between conditions. The Stressful condition in our experiment will be reworked (sound with higher arousal) so that subjects subjectively rate it higher than other conditions. Cortisol sampling is the gold standard for measuring stress, but we did not use that method. Hence, we don't have biological proof of induced stress. Future works will also use machine learning techniques to create a VRise detector based on physiological data and use questionnaires score as an index.

ACKNOWLEDGMENTS

This study was funded by European Union's Horizon 2020 RIA program under grant agreement No 883293 - INFINITY project.

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Table 1: rANOVA results of features extracted from ET, ECG, and EDA between conditions. *every 30s, **every 1 minute

Feature	p-value	f-score	effect-size
Control vs DualTask			
min_pupil_left*	0.008	39.73	0.93
sd_pupil_right*	0.019	20.97	0.86
mean_blink_rate*	0.012	30.69	0.91
TINN*	0.015	25.12	0.89
HTI*	0.044	11.20	0.79
HTI**	0.003	84.29	0.97
Control vs Stressful			
mean_acc_x**	< 0.001	857.65	0.99
mean_saccade_amplitude*	< 0.001	219.59	0.98
mean_acc_y**	< 0.001	87.55	0.96
sd_gaze_x**	0.026	12.02	0.75
TINN**	0.012	19.35	0.83
scl_mean**	0.011	19.94	0.83
Control vs Stereoscapy			
sd_pupil_left*	0.047	19.76	0.91
sd_gaze_y**	0.018	55.09	0.96
mean_distance_pupils**	0.028	33.67	0.94
DualTask vs Stressful			
min_pupil_right**	0.008	24.14	0.86
HRV_VHF**	0.029	11.11	0.74
scr_rise_sd*	0.004	34.92	0.89
scr_recovery_mean*	0.026	11.89	0.75
scr_recovery_mean**	0.006	28.46	0.88
scr_recovery_sd*	0.033	10.31	0.72
scr_recovery_max*	0.036	9.57	0.71
scr_recovery_max**	0.038	9.35	0.70
scr_recovery_sd**	0.040	8.99	0.69
scr_sd*	0.042	8.67	0.68
DualTask vs Stereoscapy			
max_pupil_left*	0.039	12.33	0.80
max_pupil_right*	0.036	13.01	0.81
mean_velocity_y*	0.003	74.85	0.96
mean_saccade_peak_velocity*	0.021	19.55	0.87
scl_min*	< 0.001	36.44	0.92
scl_min**	0.042	11.64	0.79
TINN**	0.023	18.49	0.86
HTI*	0.008	38.78	0.93
HTI**	< 0.001	2326.39	0.99
Stressful vs Stereoscapy			
min_pupil_right**	0.018	14.77	0.79
mean_saccade_amplitude*	< 0.001	253.52	0.98
mean_saccade_rate**	0.023	12.79	0.76
mean_velocity_x**	0.012	18.91	0.83
NN_sd*	< 0.001	122.93	0.97
NN_cv*	0.002	49.89	0.93
NN_number*	0.018	14.84	0.79
TINN**	0.009	21.60	0.84
scr_recovery_mean*	0.003	40.59	0.91
scr_amplitude_min*	0.005	31.85	0.89
scr_amplitude_mean**	0.038	9.29	0.69
scl_max**	0.005	30.98	0.89
scl_mean**	0.034	9.97	0.71
scl_sd**	0.051	7.63	0.66